

3	(a)	(i) Un-polarised light consists of light waves, which are transverse waves, with different planes of oscillation. Polarisation is the phenomenon in which the vibrations giving rise to waves are <u>restricted in a direction of vibration, perpendicular to the direction of wave propagation.</u>	B1 B1
		(ii) Constructive interference occurs when the superposition of two waves produce a resultant wave of greater amplitude.	B1 B1
	(b)	After P#1, wave is polarised to this axis *Angle between P#1 and #2 = $40 - 20 = 20^\circ$ After Polariser 1 (#1): $I_m = I_i \cos^2 20$ After Polariser 2 (#2): $I_f = I_m \cos^2 20$ $= (I_i \cos^2 20)(\cos^2 20)$ $= 0.780 I_i$	C1 A1
	(c)	(i) $\sin \theta = \frac{\lambda}{b}$ $\sin \theta = \frac{0.003}{2.4} = \frac{\lambda}{b}$ $b = 5.04 \times 10^{-4} \text{ m} = 0.504 \text{ mm}$ Allow range of x from 0.003 to 0.0035 m	C1 A1
		(ii) $b \downarrow$, $\sin \theta \uparrow$, $\theta \uparrow$ increase in width of central maximum, effect: (either) maxima are farther, (or) therefore the bright fringes have lower intensity (to get the second mark)	B1 B1